INCH - POUND

MIL-M-47084A(MI)
30 October 1990
SUPERSEDING
MIL-M-47084 (MI)
10 May 1974

MILITARY SPECIFICATION

MORTAR, SODIUM SILICATE, HEAT RESISTANT

This specification is approved for use by the U.S. Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements of a chemically setting sodium silicate heat resistant mortar capable of withstanding 1093° Celsius (C) for short time exposure.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5270 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 9350
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SPECIFICATIONS

FEDERAL

TT-T-548	-	Toluene, Technical

MILITARY

MIL-P-116	-	Preservation, Methods of

STANDARDS

MILITARY

MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-1190	-	Minimum Guidelines for Level C Preservation, Packing and Marking

(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Ave., Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM C 307	-	Tensile Strength of Chemical-Resistant Mortar, Grouts, and Monolithic Surfacings, Standard Test Method for
ASTM C 321	-	Bond Strength of Chemical-Resistant Mortars, Standard Test Method for
ASTM C 413	-	Absorption of Chemical-Resistant Mortars, Grouts, and Monolithic Surfacings, Standard Test Method for
ASTM C 414	-	Working, Setting, and Curing Times of Chemically Setting Chemical-Resistant Silicate and Silica Mortars, Standard Test Method for
ASTM C 579	-	Compression Strength of Chemical-Resistant Mortars, Grouts, and Monolithic Surfaces, Standard Test Methods for

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ASTM C 580	-	Flexural Strength and Modulus of Elasticity of Chemical-Resistant Mortars, Grouts, and Monolithic Surfacings, Standard Test Method for
ASTM D 2440	-	Rubber Property - Durometer Hardness, Standard Test Method for

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article sample. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.4.

3.2 Material. The mortar shall consist of a sodium silicate formulation composed of a powder filler and liquid binder capable of chemically curing at room temperature.

3.3 Mechanical properties. The mechanical properties of the cured mortar shall be as specified in table I (see 4.6.2).

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TABLE I. Mechanical properties of cured mortar.

Mechanical properties	
Property	Requirement
Tensile strength, pounds per square inch (psi), minimum	350
Flexural strength, psi, minimum	450
Compressive strength, psi, minimum	3400
Bond, strength, psi, minimum	70

3.4 Physical properties.

3.4.1 Absorption. The increase in weight of the cured mortar when boiled in toluene in accordance with TT-T-548 for not less than 2 hours shall be not greater than 15 percent (see 4.6.3.1).

3.4.2 Hardness. The shore D hardness of the cured mortar shall be not less than 70 (see 4.6.3.2).

3.4.3 Working time. The working time of the mixed mortar shall be not less than 15 minutes at 23 ± 3°C (see 4.6.3.3).

3.4.4 Heat resistance. The cured mortar shall not ignite or exhibit evidence of spalling or cracking when subjected to 1093 ± 10°C for not less than 5 minutes (see 4.6.3.4).

3.4.5 Shelf life. The shelf life of the filler and binder in unopened containers shall be not less than 1 year from date of manufacture when stored within the temperature range of 1.7 to 26.7°C (see 4.6.3.5).

3.5 Workmanship. The workmanship shall be such as to insure a product which is uniform and in accordance with the requirements of this specification (see 4.6.3.6).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the

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performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.6.

4.4 First article inspection. When specified in the contract or order (see 6.2) a first article sample shall be subjected to first article inspection. The quantity shall be as specified in the contract or order. First article inspection shall consist of all examinations and tests specified herein. Subsequent units shall not be considered for acceptance until Government approval of the first article sample has been obtained.

4.5 Quality conformance inspection. Quality conformance inspections shall be as specified in table II.

TABLE II. Quality conformance inspection.

Inspection	Requirement paragraph	Test paragraph
Hardness	3.4.2	4.6.3.2
Working time	3.4.3	4.6.3.3
Heat resistance	3.4.4	4.6.3.4

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4.5.1 Lot size. Lot size shall consist of all the filler and binder manufactured at one time, from one batch, forming a part of one contract or order, and submitted for acceptance at the same time and place.

4.5.2 Sampling. Unless otherwise specified (see 6.2), 1 pint of filler and 0.25 pint of binder from one unit representative of each lot shall be selected at random for quality conformance testing. Each container of filler and binder shall be considered as a unit of product. Failure of the sample to meet the quality conformance requirements shall be cause for lot rejection.

4.6 Test methods.

4.6.1 Test specimen preparation. The applicable test specimens shall be prepared by mixing the filler and binder in accordance with the manufacturer's recommendations. The specimens shall be cured for 24 to 48 hours at 25 ± 3°C.

4.6.2 Mechanical property tests.

4.6.2.1 Tensile strength. The tensile strength shall be determined in accordance with ASTM C 307 and conforming to table I.

4.6.2.2 Flexural strength. The flexural strength shall be determined in accordance with ASTM C 580 and conforming to table I.

4.6.2.3 Compressive strength. The compressive strength shall be determined in accordance with ASTM C 579 and conforming to table I.

4.6.2.4 Bond strength. The bond strength shall be determined in accordance with ASTM C 321 and conforming to table I.

4.6.3 Physical property tests.

4.6.3.1 Absorption. The percentage of weight increase shall be determined in accordance with ASTM C 413 and conforming to 3.4.1.

4.6.3.2 Hardness. The shore D hardness shall be determined in accordance with ASTM D 2440 and conforming to 3.4.2.

4.6.3.3 Working time. The working time of the mixed mortar shall be determined in accordance with ASTM C 414 and at the working time and temperature specified in 3.4.3.

4.6.3.4 Heat resistance. Three cubes of cured mortar measuring 0.25 x 0.25 x 0.25 inch shall be cut from the specimen with a suitable instrument and shall be within a tolerance of ± 0.03 inch. Each cut specimen shall be individually placed into a muffle furnace heated to 1093 ± 10°C. The specimen shall be elevated approximately 2 inches off the floor of the furnace, but not farther than 4 inches from the thermocouple, by means of a fire brick. The

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specimens shall not ignite after exposure for not less than 5 minutes at the temperature specified in 3.4.4. The specimens shall not exhibit evidence of spalling or cracking.

4.6.3.5 Shelf life. The filler and binder shall be tested for conformance with all the requirements specified herein after 1 year of storage at the temperature specified in 3.4.5. The supplier's written certification of compliance with 3.4.5 may be accepted during the storage period or as specified in the contract or order.

4.6.3.6 Workmanship. The material shall be visually examined to assure conformance to the workmanship requirement specified in 3.5.

4.7 Inspection of packaging. Except when commercial packaging is specified, the sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification shown in section 5. The inspection of marking for shipment and storage shall be in accordance with MIL-STD-129. The inspection of commercial packaging shall be as specified in the contract or order (see 6.2).

5. PACKAGING

5.1 Preservation and packing. Level C packaging (see 6.2).

5.2 Level C. Unless otherwise specified in the contract or order, preservation and packing shall be in accordance with MIL-STD-1190.

5.3 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This specification covers the requirements of a chemically setting sodium silicate mortar capable of withstanding high temperature for short time exposure.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

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c. When first article is required (see 3.1 and 4.4).

d. First article sample quantity (see 4.4).

e. Sampling plan if other than specified (see 4.5.2).

f. Requirements for certification of storage life (see 3.4.5 and 4.6.3.5).

g. Requirements for inspection of commercial packaging (see 4.7).

h. Requirement for packing and marking (see 4.7 and section 5).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerers whether the item(s) should be a first article sample, a first production item, or a number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Metrication. Metric equivalents in accordance with FED-STD-376 are acceptable for use in this specification.

6.5 Subject term (keyword) listing.

Chemical compound
Caulking material

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - MI

Review Activity
DLA-GS

Preparing Activity:
Army - MI

(Project 9350-A106)

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-M-47084A	30 October 1990

3. DOCUMENT TITLE
MORTAR, SODIUM SILICATE, HEAT RESISTANT

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER		
a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	7. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY	
a. NAME COMMANDER U.S. ARMY MISSILE COMMAND	b. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON (205) 876-6980 or 746-6980
c. ADDRESS *(Include Zip Code)* ATTN: AMSMI-RD-SE-TD-ST Redstone Arsenal, AL 35898-5270	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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